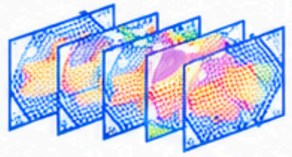
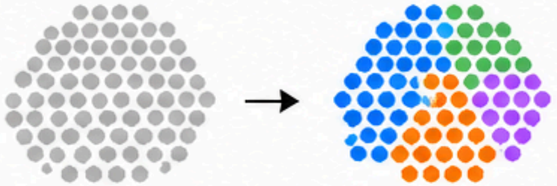
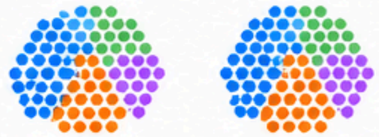
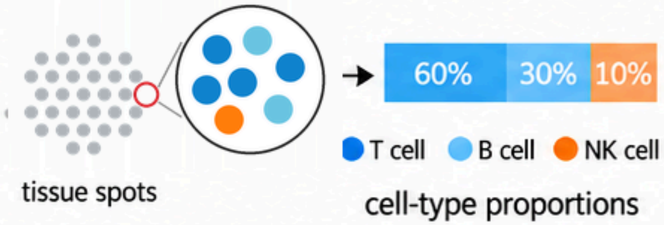
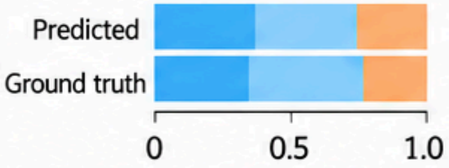
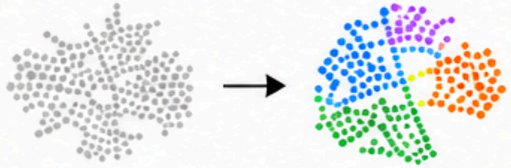
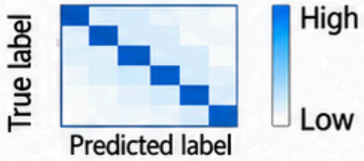
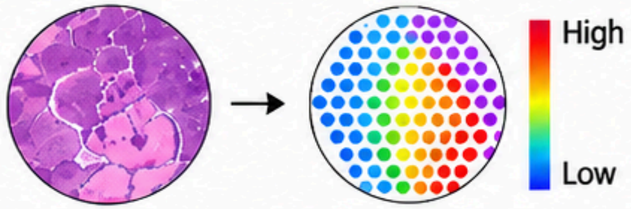
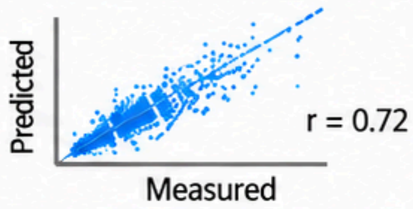
b

Overview of ST-Bench

	Input datasets	Task	Benchmark protocol	Evaluation metrics
1 Spatial clustering	 5 datasets spot- & single-cell resolution	 spots → spatial domains	9 FMs (zero-shot + fine-tuned) 3 specialized methods ▼ Leiden clustering on embeddings + spatial smoothing	ARI / NMI  Prediction Ground truth
2 Deconvolution	 7 datasets real & simulated, with / without ground truth	 tissue spots → 60% 30% 10% T cell B cell NK cell cell-type proportions	9 FMs (zero-shot) 3 specialized methods ▼ cosine-softmax / NNLS mapping to scRNA-seq reference	PCC / JSD / RMSE  Predicted Ground truth 0 0.5 1.0
3 Cell-type annotation	 2 datasets Xenium & MERFISH	 unlabeled → labeled	7 FMs (zero-shot) ▼ kNN label transfer from annotated reference	Accuracy / Macro-F1  True label Predicted label High Low
4 Gene expression prediction from H&E	 6 Visium datasets with paired H&E images	 H&E → Gene X expression High Low	OmiCLIP · STPath 2 specialized methods ▼ official zero-shot prediction / ridge probe on H&E embeddings	Pearson r  Predicted Measured r = 0.72